

ALBARTVILLE, AL, USA

WMO: 720376

Lat: 34.229N

Lon: 86.256W

Elev: 1032

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	24.7	6.8	8.1	28.8	10.1	9.6	30.5	22.4	39.5	19.7	43.2	7.5	350	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	93.4	73.0	91.0	73.9	89.9	73.7	80.3	86.1	79.1	84.9	77.5	83.1	6.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	156.3	83.5	77.3	147.6	82.0	76.6	144.3	81.4	44.8	87.8	43.3	84.2	42.0	83.4	97.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	16.0	13.2	DB	15.4	96.8	5.0	3.4	11.8	99.3	8.9	101.3	6.1	103.2	2.4	105.7	
			WB	13.2	83.0	4.2	5.2	10.1	86.8	7.7	89.8	5.3	92.7	2.2	96.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	41.1	45.5	54.4	62.0	69.5	76.9	78.7	78.6	73.8	62.4	51.3	46.4
	DBStd	15.25	10.29	9.74	9.85	7.69	6.46	3.85	3.50	4.18	5.55	8.12	8.20	9.44
	HDD50	855	311	187	72	9	0	0	0	0	0	7	81	188
	HDD65	2961	740	547	344	144	36	0	0	0	7	148	415	580
	CDD50	5163	37	61	209	369	605	807	891	885	714	391	119	75
	CDD65	1793	0	1	16	54	176	357	426	420	271	67	3	2
	CDH74	15515	0	1	95	395	1362	3283	3857	3851	2162	492	15	2
	CDH80	5912	0	0	7	52	372	1343	1605	1656	771	105	1	0
(14) Wind	WSAvg	5.5	7.0	7.2	7.1	6.5	5.0	4.3	3.8	3.8	4.1	5.4	6.0	6.5
(15) Precipitation	PrecAvg	55.2	5.6	5.2	5.7	4.3	4.4	4.4	4.3	3.9	3.9	3.2	4.5	5.1
	PrecMax	69.2	9.1	10.3	9.5	7.6	10.4	10.1	7.6	9.1	8.6	9.3	10.3	10.0
	PrecMin	35.0	2.1	2.3	1.3	2.3	0.3	1.3	1.5	1.5	0.1	0.0	2.4	2.4
	PrecStd	8.0	1.7	2.0	1.9	1.4	2.5	2.4	1.9	1.7	2.3	2.4	1.9	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.5	72.3	81.2	84.5	89.8	95.1	96.8	98.9	92.7	87.8	75.6	72.0
		MCWB	59.5	62.4	62.7	64.3	70.0	72.0	72.7	72.4	69.9	68.9	63.1	67.3
	2%	DB	63.5	68.1	77.0	81.3	86.4	91.2	92.7	94.7	90.4	82.1	71.8	68.0
		MCWB	58.2	60.3	61.7	64.9	69.9	72.9	75.2	73.5	71.3	67.7	60.1	63.8
	5%	DB	61.0	64.0	72.5	79.0	84.1	90.2	90.4	90.8	88.0	79.3	68.2	64.0
		MCWB	57.2	56.2	60.6	64.3	69.5	72.9	75.6	74.4	72.3	65.9	60.1	61.1
	10%	DB	57.1	61.2	69.8	75.2	81.5	88.0	88.2	88.4	84.4	76.6	64.2	61.3
		MCWB	53.3	53.9	58.7	62.9	68.5	72.8	75.2	74.1	71.7	64.8	58.0	58.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	65.2	67.5	72.4	75.9	80.5	82.0	81.0	79.0	76.2	70.0	69.5
		MCDB	64.1	69.9	72.2	78.7	81.4	87.2	88.1	85.9	84.8	80.6	71.5	70.4
	2%	WB	60.7	63.2	65.1	69.4	73.9	78.9	80.6	79.7	77.4	73.1	66.1	64.9
		MCDB	61.9	65.9	71.2	75.0	80.4	85.9	86.5	85.0	83.4	77.5	68.4	66.1
	5%	WB	58.4	60.0	63.3	66.9	72.4	77.3	79.6	78.3	75.9	71.1	63.1	62.5
		MCDB	60.9	62.4	68.1	74.5	79.5	83.8	85.4	83.8	82.6	76.3	64.6	63.7
	10%	WB	54.4	55.8	61.3	65.0	71.0	75.5	77.7	77.0	74.1	68.3	60.2	58.9
		MCDB	56.8	59.1	67.1	72.4	78.2	82.9	82.6	83.1	80.1	73.8	63.5	61.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	17.0	18.6	20.6	19.2	19.4	18.0	18.4	18.8	19.9	19.4	15.4
		MCDBR	18.2	20.8	21.6	22.2	22.2	22.5	20.9	21.8	22.0	22.7	21.6	17.4
		MCWBR	13.8	14.0	11.5	10.0	8.6	7.6	7.5	7.0	8.0	10.7	13.1	14.4
	5% WB	MCDBR	15.3	16.6	18.1	18.9	17.8	18.6	18.2	17.6	17.8	18.0	16.1	16.4
		MCWBR	13.7	14.2	11.5	10.6	9.1	9.9	10.2	9.2	9.3	10.9	13.4	15.4
(40) Clear-Sky Solar Irradiance	taub		0.311	0.315	0.355	0.385	0.439	0.471	0.517	0.517	0.421	0.369	0.334	0.313
	taud		2.518	2.503	2.411	2.313	2.205	2.149	2.042	2.028	2.288	2.423	2.462	2.528
	Ebn at Noon		286	297	291	286	270	260	248	245	266	273	274	279
	Edh at Noon		26	30	35	41	46	49	54	54	39	32	28	25
(44) All-Sky Solar Radiation	RadAvg		800	965	1330	1722	1883	1965	1879	1745	1513	1244	909	682
	RadStd		59	123	123	117	149	153	115	94	136	160	99	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+203

Nomenclature: See separate page